Final Report
Moonbase AndrewMills-1
Submitted for SPC 5090

Andrew Mills
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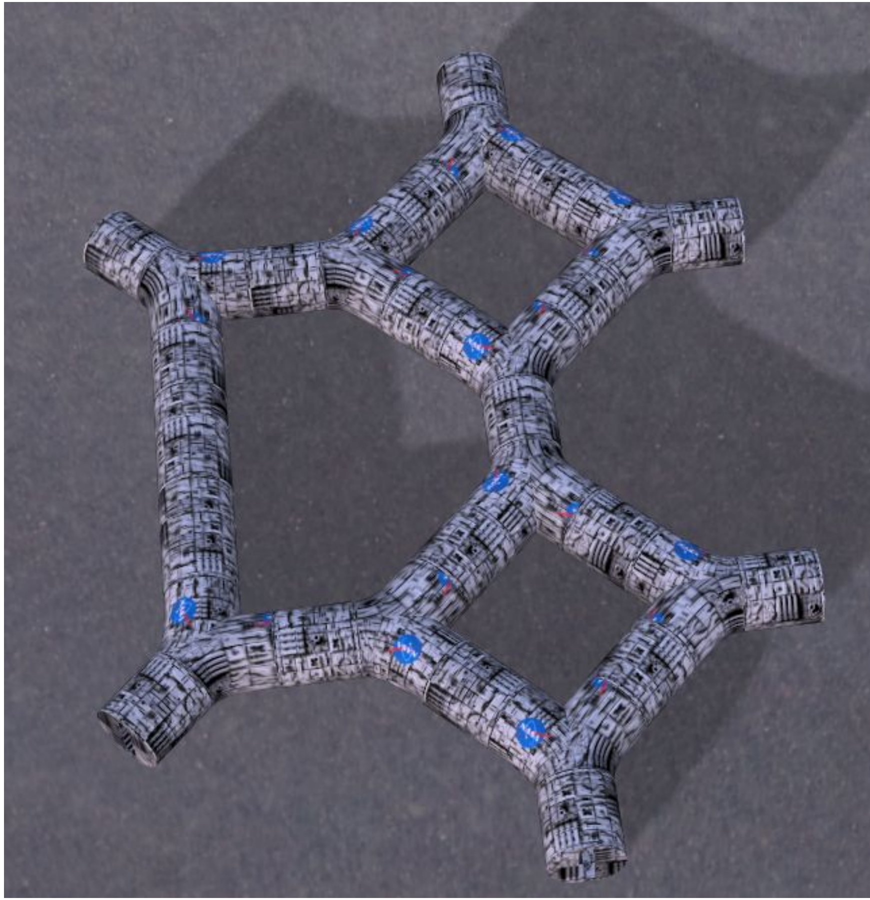


1.) Summary

Moonbase AndrewMills-1 is designed for:

- Crew: 6
- Mission length: 180 days
- Location: Near side, close to the equator within Mare Smythii
- Purpose:
  - Base operations demonstration with surface exploration
  - ISRU demonstration
  - Highlands/Mare selenology field study
  - Lunar surface-based astronomy
  - Low-gravity biological and agricultural research

2.) Lunar environment affecting the Moon

- 14 days of sun, 14 days of night
- High surface temperature variations
- Radiation
  - Galactic Cosmic Rays (GCR)
  - Solar Particle Events (SPE)
  - Solar wind
- Meteoroids
- Moon dust
- The unknown

3.) Science

There will be 4 science modules connected to the base:

- Biology lab
  - For study of how microbes, plants, animals, and humans react to and change within the lunar environment.
- Greenhouse and hydroponics modules
  - For study of plant growth in a low gravity environment, and to provide fresh food for the astronauts.
- Selenology lab
  - For study of lunar surface samples.
- Astronomy module
  - For study of the stars from an atmosphere-less environment.

4.) Location

Moonbase AndrewMills-1 is located at coordinates -0.932 °N, 81.217 °E within Mare Smythii (figure 1), adjacent to Avery crater (figure 2).

Figure 1: AndrewMills-1 on lunar surface

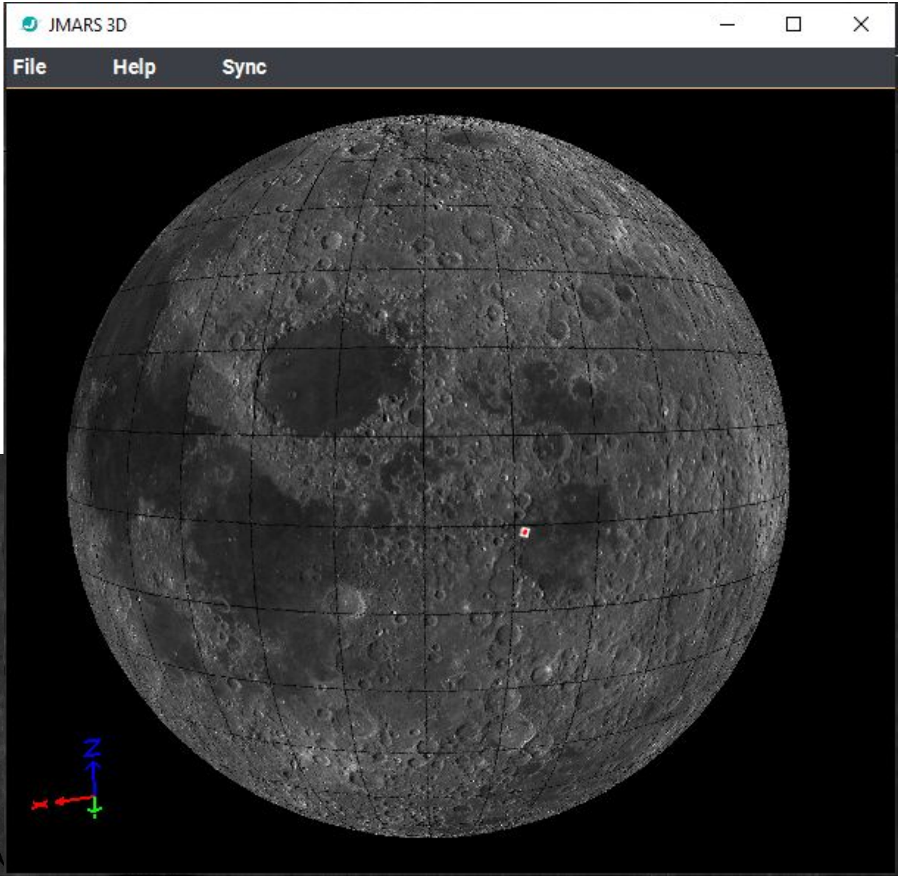
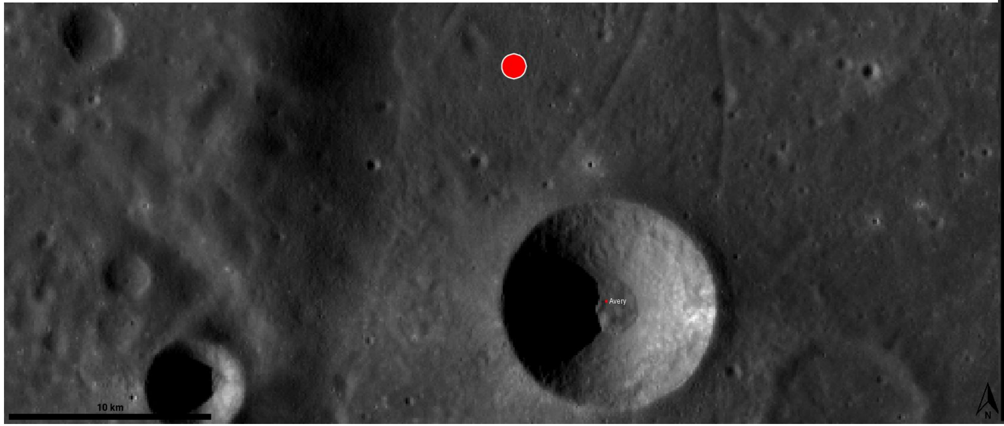


Figure 2: AndrewMills-1 site at 10 km scale on lunar surface



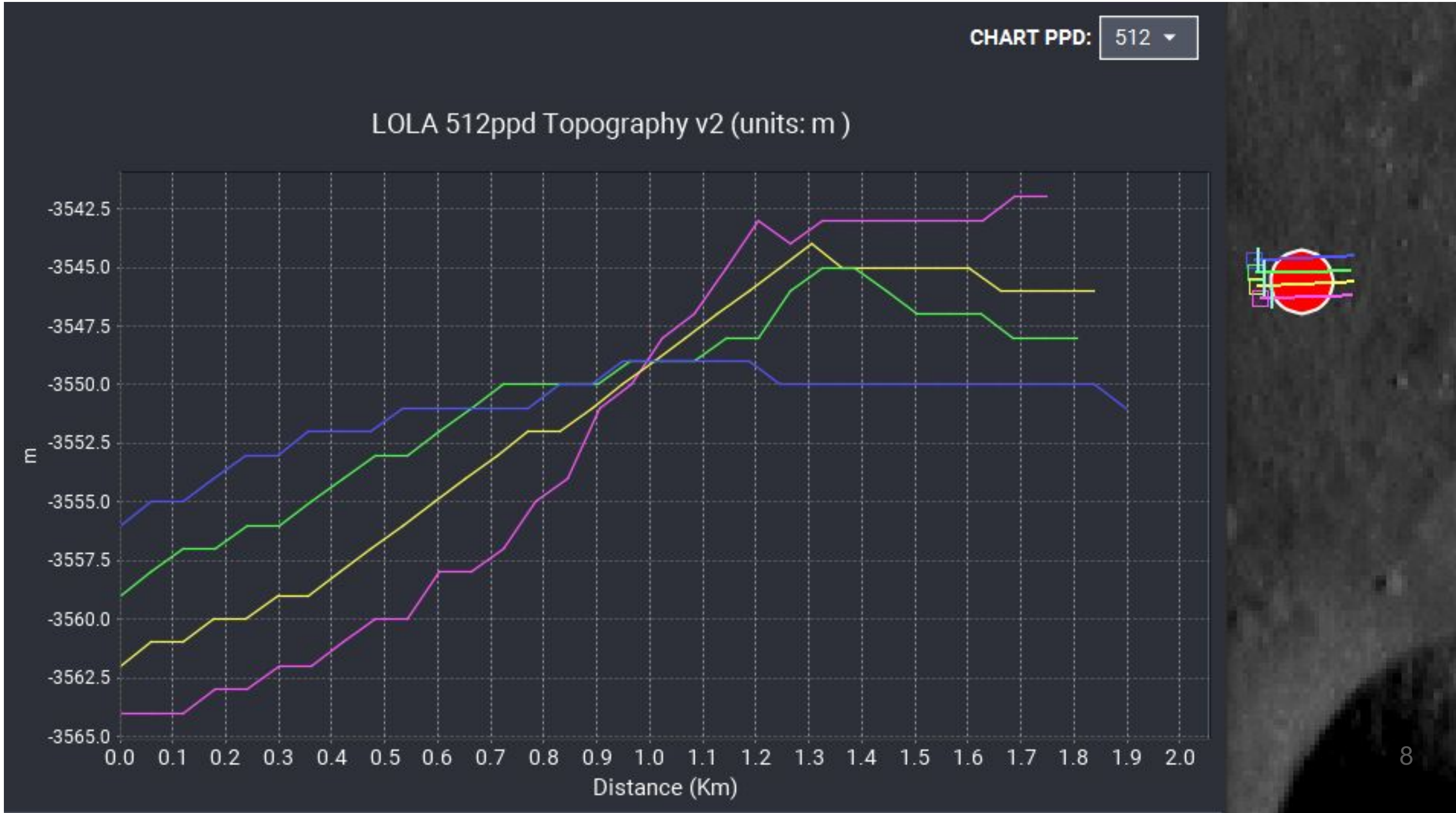
4.) Location requirements

Along with the course requirements for a location on the near side close to the equator, I pulled siting requirements from the “Site Selection Strategy for a Lunar Outpost” workshop:

- Within 10° latitude of the lunar equator to maximize astronomical observation access to the celestial sphere, and within 2° of the lunar equator for space physics observations of the Sun. (-0.932 °N)
- Within 5-10° longitude of the mean lunar limb to maintain Earth in constant FOV low on the horizon. (81.217 °E, mean lunar limb located at 90 °E)
- At a mare-highland contact on mare surface for selenological exploration. (13.6 km to nearest highland surface)
- Should be located within a flat area for instrument emplacement, and should allow placement at distances of several kilometers away to minimize interference. (~23 m of elevation change over 1km around site, Figure 3)

4.) Location requirements continued

Figure 3: Elevation profiles along selected site in meters.



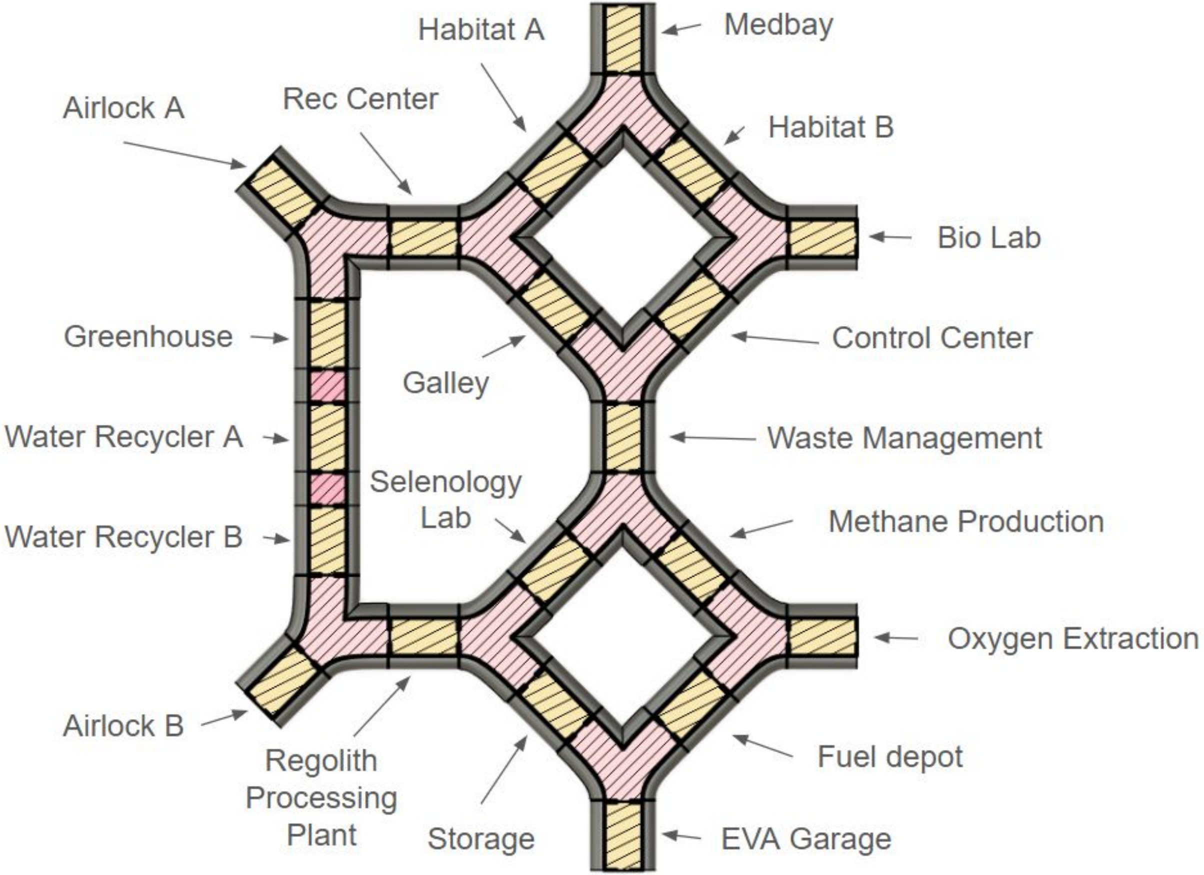
4.) Location verification

The Avery crater location meets these requirements, and was chosen because of:

- Communication with Earth low on the horizon in constant LOS
- Selenological interest in the Mare/highland contact region
- Flat areas for easy base construction and instrument placement

5.) Elements of the Moon base

Figure 4:
Base layout



5.) Elements of the Moon base continued

- There are a total of 20 functional modules.
- The modules are separated into three wings, North, South, and West.
- There is a mixed configuration of radial and courtyard for the North and South wings, and a linear configuration for the West wing.
- The three wings are interconnected, allowing for two entry and exit ways through each module to an airlock.
- Each functional module is connected by joint modules, allowing for 90 degree, 135 degree, and axial connections.
- The South wing will be a “dirty” wing, where lunar dust should be expected and PPE should be worn. Dust-removal systems will be available at the intersections between the south module and other modules.
- There are 10 Y-Joint modules (135-135-90), and 2 I-Joint modules (axial).

6.) Habitats and interiors

The following figures illustrate the floor plans of each module. Each functional module (figure 5) has 5 m by 2.75 m of floor space, with 3.5 m of height. 1 m of walkway space is always available in every module. Each doorway is 1 m by 3.5 m, with a 0.25 m space for a sliding hatch door. Modules are connected using I-joint (figure 6) or Y-Joint (figure 7) connectors. Joint modules will be lined with hydroponic bays to provide food and greenery for the astronauts.

Figure 5: Functional module

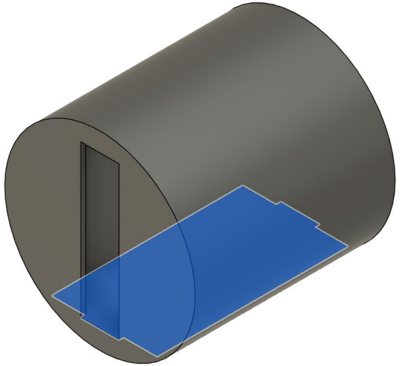


Figure 6:
I-Joint

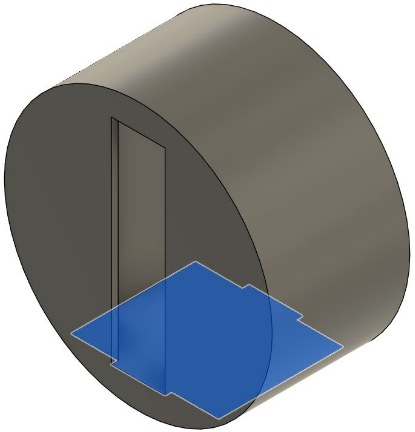
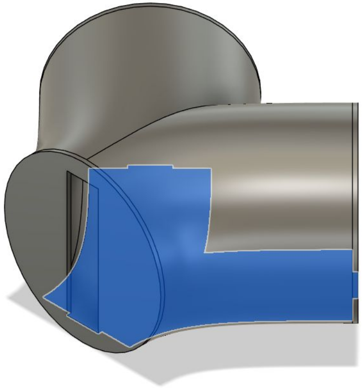


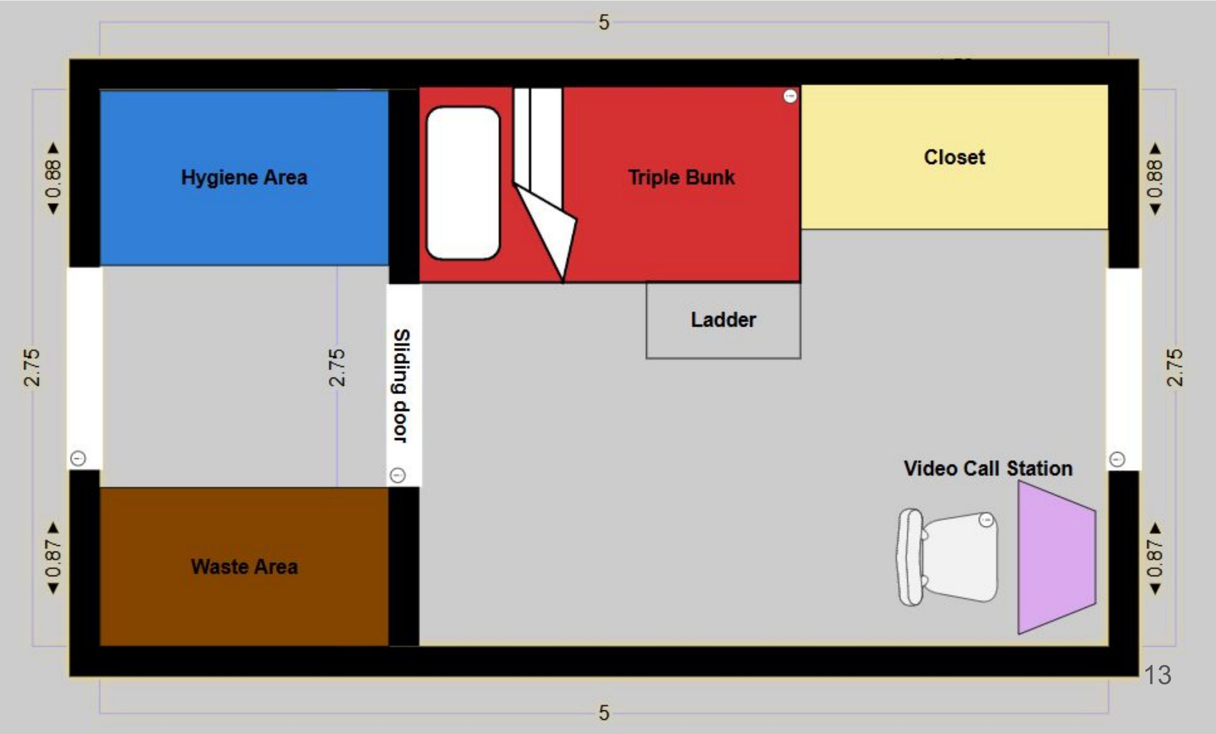
Figure 7:



6.) Habitats and interiors: Crew quarters (figure 8)

- Triple-twin bunk with ladder access.
- Separate bathroom with hygiene and waste areas.
- Closet for everyday clothes and personal items.
- Video call/work desk for constant family contact.

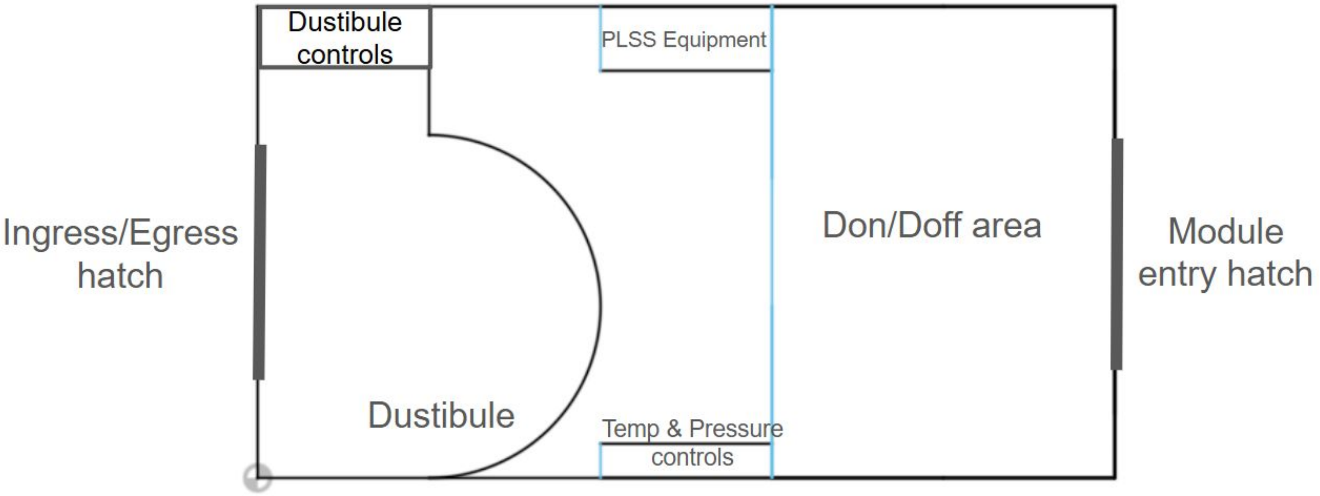
Figure 8: Crew quarters



6.) Airlock module (figure 9)

- 4 EVA suits will be stored just outside of each airlock in the adjacent joint module.
- “Dustibule” at the entrance hatch.
- Temperature and pressure control panel for airlock control.
- PLSS canisters and batteries will be stored.

Figure 9: Airlock layout



6.) Habitats and interiors: Volumes & Areas

Free floor area was estimated by using the floorplan tool SmartDraw to find the unoccupied area in each module, then by using Fusion 360 to estimate the free area in each joint and adding them together. The estimated total free floor area comes out to 398.479 m^2 as shown in table 1.

Free volume was estimated by taking the free floor area and multiplying it by the ground-to-ceiling height. The estimated total free volume comes out to 1394.677 m^3 as shown in table 1.

These free floor space and volume numbers meet NASA’s standards for 6 crew for 6 months.

6.) Habitats and interiors: Volumes & Areas continued

Table 1: Module area and volume list

Module	Free floor area (m2)	Free volume (m3)
Habitat (2)	6.88	24.08
Lab (2)	8.82	30.87
Production (6)	7.23	25.305
Medbay	7.62	26.67
Rec Center	10.62	37.17
C&C	8.88	31.08
Galley	8.61	30.135
Storage (3)	5	17.5
Airlock (2)	13.175	46.1125
I Joint	6.625	23.1875
Y joint	28.343	99.2005
Total	398.479	1394.6765

7.) Thermal control: The external environment

- During lunar day, surface temperatures vary from a min of 120 K to a max of 373.89 K along the curve shown in figure 10 as a function of sun angle.
- During lunar night, a constant surface temperature of 120 K is maintained.
- Components of heat transfer:
  - Solar radiation
  - Space background radiation
  - Sunlit surface reflection
  - Regolith heat conduction

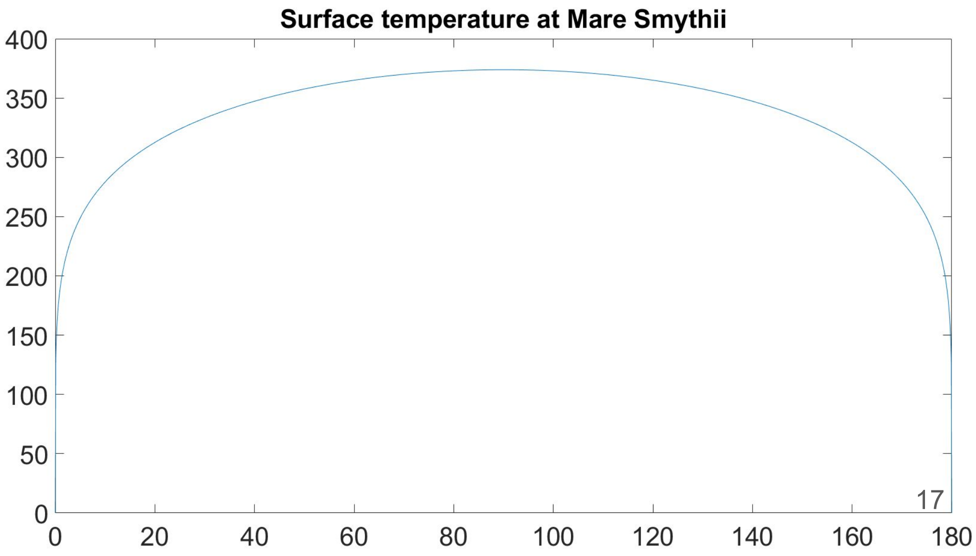


Figure 10: Surface temperature (K) vs. sun angle (deg) over a lunar day

7.) Thermal control: The internal environment

Ideal temperature depends on the function of the module, but all should be maintained at a comfortable shirtsleeve temperature for the sake of astronaut comfort. Estimates for hourly heat output of major base systems (based on Earth analogs) are:

- Astronauts: 100 W/hr each
- Personal computers: 350 W/hr each
- Health monitors: 100 W/hr each
- LED lighting: 30 W/hr each
- Production: 2000 W/hr each machine (Could be higher)

8.) Thermal control: Active control

The active thermal control system will consist of three major components:

- Cold plates
- Heat pipes
- Radiators

Electrical and production components will interface with cold plates where heat pipes will carry a working fluid to and through deep-space facing radiators and back as shown in figure 11.

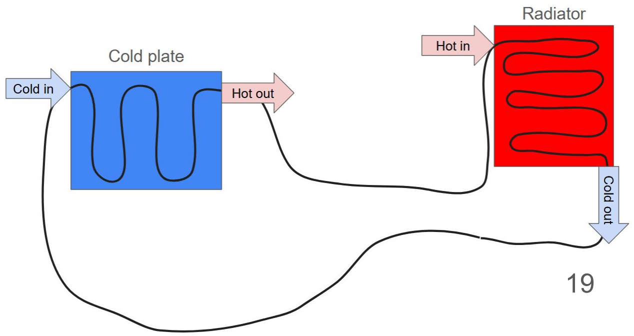
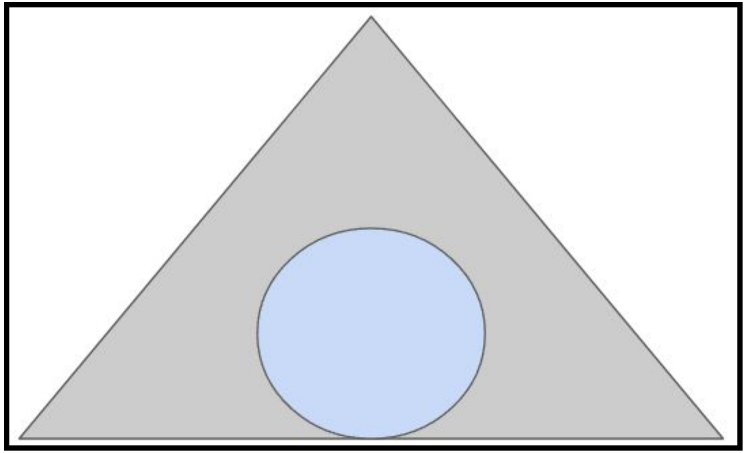


Figure 11: Cold plate-radiator system

7.) Thermal control: Passive control

A passive thermal control system made up of a 2 m regolith layer shown in figure 12 will protect the crew from the massive temperature swings of the lunar environment.

Figure 12: Module (blue) covered by lunar regolith (grey)



8.) Environment hazards: Radiation

- Serious issue, needs mitigation for safe astronaut activities. Made up of two major types:
  - Solar Particle Events (SPEs) - Rare, intense bursts of medium energy protons
  - Galactic Cosmic Rays (GCRs) - Constant stream of high energy ions
- Dose equivalent can be determined using the “Q”, or quality factor, times the absorbed dose. This value gives a metric to allowable radiation exposure.
- Shielding would be required to avoid reaching the maximum allowable dose equivalent for astronauts.

8.) Environment hazards: Temperature

- High lunar day temperatures could expose unprotected astronauts to heat exhaustion.
- Low lunar night temperatures could expose astronauts to frostbite and hypothermia.
- These possibilities make robust thermal control systems a requirement.

8.) Environment hazards: Meteoroids

- Meteoroids pose a penetration hazard to space operations due to their high speeds. There is a constant flux of meteoroids hitting the moon at rates of:
  - 1088.42 impacts/m^2/year of 1e-4 cm or greater
  - 1.61142e-18 impacts/m^2/year of 700 cm or greater
- This creates a requirement for shielding from meteoroid impacts to avoid penetration and decompression of base elements.

9.) Shielding concept

Regolith shielding provides an in-situ, passive, failsafe solution to mitigating the environmental effects on the lunar surface of:

- Radiation effects: brought down to acceptable levels with 1-2 m of regolith.
- Thermal effects: brought down to acceptable levels with 0.1-10 m of regolith.
- Meteoroid effects: brought down to acceptable levels with 0.5 m of regolith.

Moonbase AndrewMills-1 will use a 2 m regolith shield to meet these requirements.

9.) Regolith usage: Regolith shield

There are four possible configurations of lunar regolith shielding for Moonbase AndrewMills-1:

- Case 1: Regolith retainer structure (figure 13)
- Case 2: Unsupported minimum mound structure (figure 14)
- Case 3: Flat bin structure (figure 15)
- Case 4: Semicircle bin structure (figure 16)

Figure 15: Flat bin structure

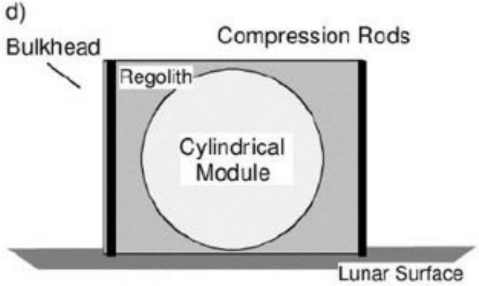


Figure 13: Regolith retaining structure

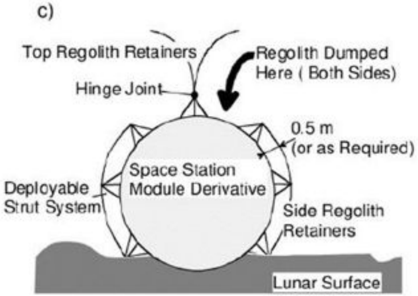


Figure 14: Unsupported minimum mound

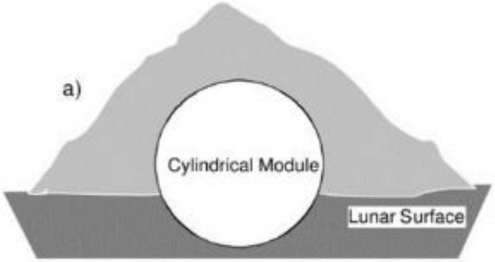
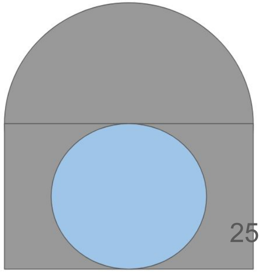


Figure 16: Semicircle bin structure



9.) Regolith usage: Regolith shield

Regolith required per module was calculated in each case to adequately shield Moonbase AndrewMills-1 (table 2), and found that case 4 would require the least regolith. Despite this, case 2, the unsupported minimum mound structure, will be chosen due to its quick setup and maintenance.

The regolith shield will be deposited over the base using an autonomous robotic system prior to astronaut arrival, and the base will continue to use these robots for regolith collection after the astronauts arrive.

Table 2: Regolith required per module in m^3

Empty module volume	98.17477042
Case 1	219.9114858
Case 2	214.245
Case 3	216.8252296
Case 4	173.3683577

10.) Power Supply and Energy Storage

Colozza’s 2020 paper suggests that power demands for a 6-person lunar base would total 53.85 kW, so the following power architecture decisions are based on this need.

- Power system architecture on Moonbase AndrewMills-1
  - Solar power
    - DOE multi-junction solar cells
  - Nuclear power
    - KRUSTY stirling reactors
  - Energy storage
    - Li-S battery array

10.) Power Supply and Energy Storage- Solar Array

2 Options for solar cells:

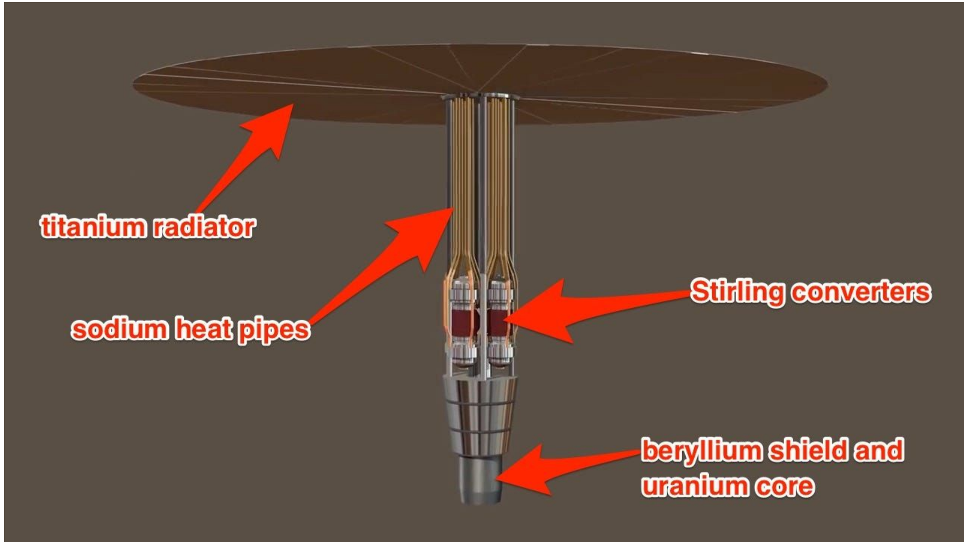
- Gallium Arsenide: More common, efficiency of 32%. Would require a 123.736m^2 solar array to meet the power needs of the base.
- DOE Multi-junction: Still in development, tested peak efficiency of 45%. Would require a 87.990m^2 solar array to meet the power needs of the base.

The second option was chosen despite the fact that it’s still in development due to the much lower array area required. The solar array will be located ~1 km SW of the base to avoid the dust plumes from launch and landing

10.) Power Supply and Energy Storage - Nuclear Reactor

KRUSTY reactors (figure 17) are Stirling generators using fission as a heat source. Tested successfully on Earth, they would be assumed to produce 10kW of power each and with a power requirement of 53.85 kW, 6 KRUSTY reactors would be required. These reactors would be located alongside the solar array.

Figure 17:
KRUSTY reactor



10.) Power Supply and Energy Storage - Batteries

Despite the benefits that come from Lithium-ion batteries, Lithium-Sulfur batteries boast a higher specific energy, leading to a reduction in mass needed for launch. Based on the power generation of the solar array and KRUSTY reactors, the total energy needing storage over a lunar day would be 38.25 MWh, necessitating 1524.05 kg of Li-S batteries.

11.) Life Support and Crew Health

Life support on Moonbase AndrewMills-1

- Atmosphere management: air will be constantly flowed through all habitats using fans, in through a vent in the ceiling and out through vents near the floor. Recollected air will be scrubbed of CO_2 and contaminants with a similar system to the ISS before being re-released.
- Water management: Two water-recycling modules in the east wing provide the means to clean used water, where a system similar to that on the ISS will provide clean water back to the astronauts.

11.) Life Support and Crew Health

- Food production and storage: Shelves in the galley and a storage module in the south wing allow storage of non-perishable food, while preparation will take place in the galley.
- Waste management: Trash and human waste will be collected, and whatever can be recycled will be. Urine can be cleaned and reused, and trash may serve other purposes, such as food waste and feces as compost.
- Medical and safety aspects: The medbay provides astronauts with medical support. Fire detectors will be embedded in the air-scrubbing system, and module hatches will be sealable to slow fire spread. Radiation shielding is available with the regolith shield.

11.) Life Support and Crew Health

Consumable needs (table 3) assuming the following:

- Considering a water recycling rate of 98% (as seen on ISS)
- Oxygen scrub/recycle rate of 50% (as seen on ISS)
- No crop grows on the lunar surface
- Needs and wastes match those shown in table 4

Table 3: Daily and total consumable needs in kg of the mission

With ISS levels of reclamation		
Consumable	Daily	6 months
Oxygen	1.95	351
Water	0.33	59.4
Food	3.9	702
Total	6.18	1112.4

11.) Life Support and Crew Health

Table 4: Daily needs and wastes of 6 crew

6 Crew			
Input	Amount	Output	Amount
Oxygen	5.25	CO2	6.6
Food prep water	3	Respiration and perspiration water	9.9
Drink	12	Urine	8.4
Water in food	1.5	Fecal water	0.6
Food solids	3.9	Waste solids	0.15
Total	25.65	Total	25.65

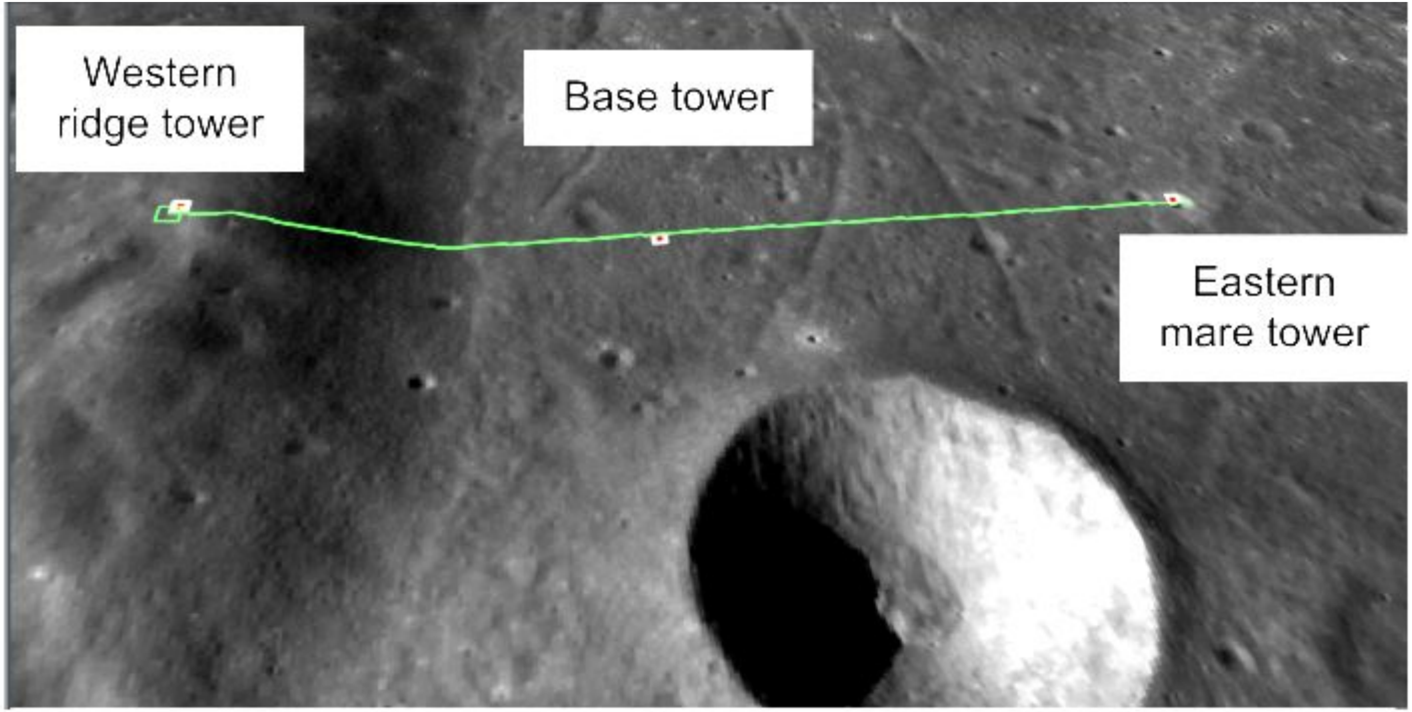
12.) Communication and Navigation Systems - Comms

Three 10m tall communications towers will cover the area surrounding the base, using the natural features nearby to its advantage. (figure 18)

- Base tower: adjacent to the base, for communication with Earth and nearby surface assets.
- Western highland tower: located 1550m above the base atop the edge of Mare Smythii, providing unobstructed high-gain LOS communications in a 73.6km radius in a 180 degree arc to its east (figure 19).
- Eastern mare tower: Located ~20km east of the base, used to transmit high-density data from further east into the mare region.

12.) Communication and Navigation Systems - Comms

Figure 18: Comm tower layout



12.) Communication and Navigation Systems - Comms

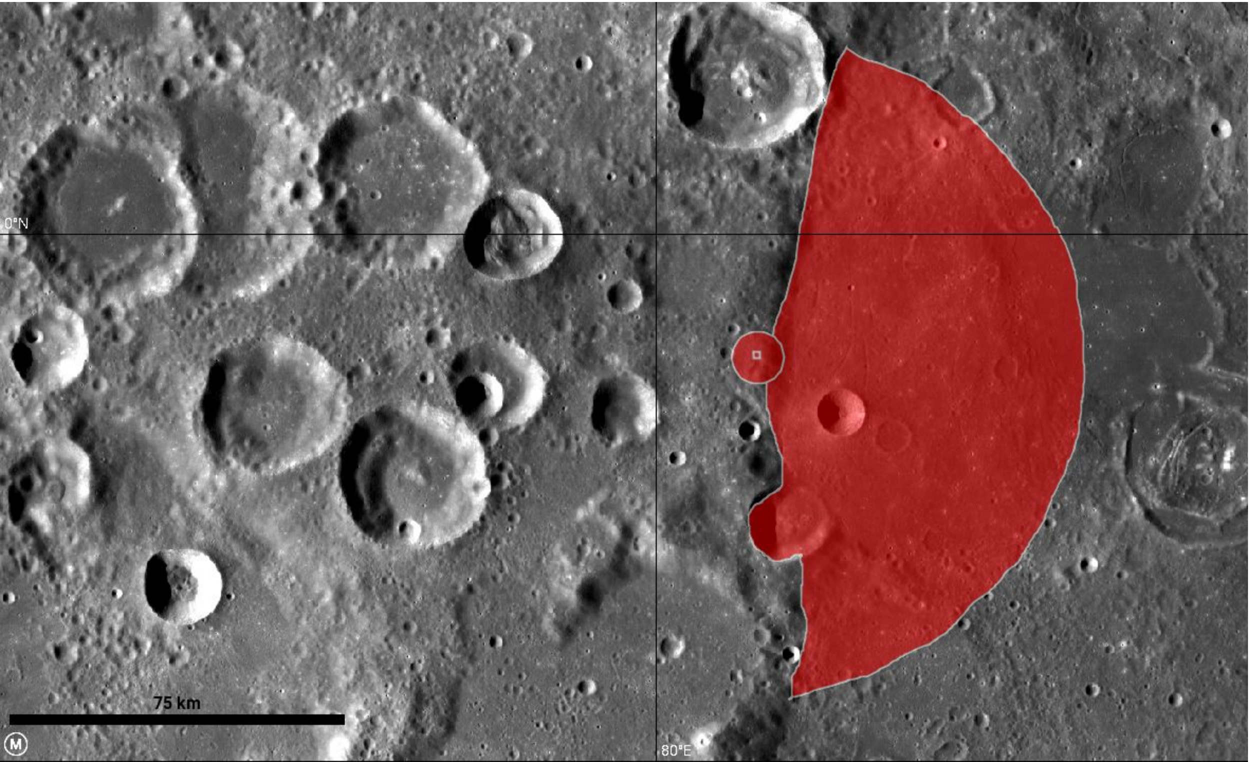


Figure 19: Western ridge tower LOS

12.) Communication and Navigation Systems - Nav

Navigation beacons will be placed at the base of each of the communications towers to provide fixed points for exploration systems to navigate from. LOS with Earth along with one or two navigation beacons should provide accurate positioning within the base area.

13.) EVAs and Lunar Surface Vehicles

There are three types of EVA expected of moonbase astronauts:

1. Scheduled EVA: Planned before the mission and included in nominal mission timeline.
2. Unscheduled EVA: Not included in nominal timeline but may be necessary to achieve success.
3. Contingency EVA: Required for safety of the crew, such as evacuation or emergency return to the habitat.

13.) EVAs and Lunar Surface Vehicles

Surface vehicles:

- 2 unpressurized LTVs (figure 20): Lunar Outpost’s design for a modular rover. Incapable of operation at night. Provides:
  - Personnel/hardware transport.
  - Production material import.
  - Surface excursions
- 1 pressurized Lunar Cruiser LTV (figure 21): JAXA’s design for a long range pressurized rover. Capable of providing:
  - Long range surface transport.
  - Secondary habitat

13.) EVAs and Lunar Surface Vehicles

Figure 20: Lunar Outpost unpressurized Rover

Figure 21: JAXA Lunar Cruiser pressurized rover

14.) Launch/Landing Vehicles and Site

- Launch/landing site 2 km NW of the base (figure 22) at 0.864 deg S, 81.196 deg E, and will support the use and partial refuelling of the SpaceX Starship (figure 23)
- Berms will be piled alongside the site using the robotic excavators used to bury the base.
- 2 km away was chosen for landing dust mitigation purposes, but can’t be any further since 2 km is considered a walking EVA upper limit.

14.) Launch/Landing Vehicles and Site

Figure 21: Launch/landing location (blue) next to base location (red)

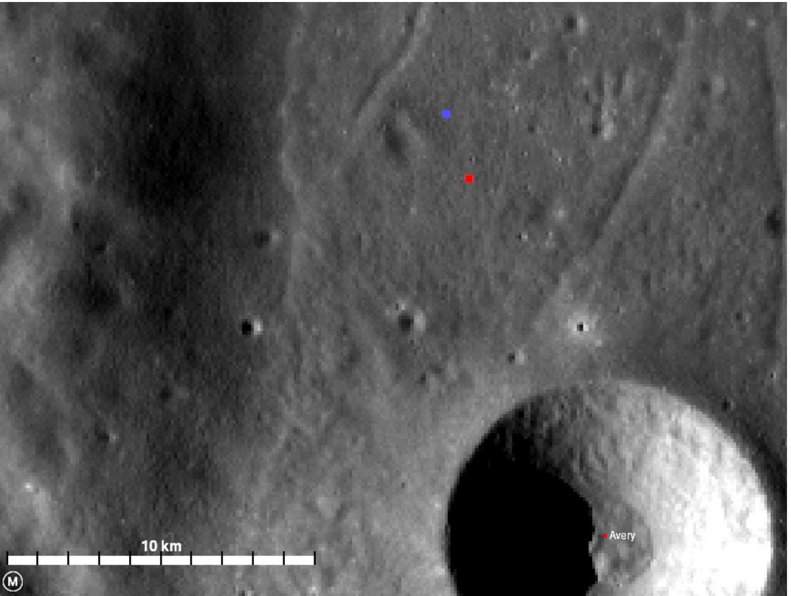


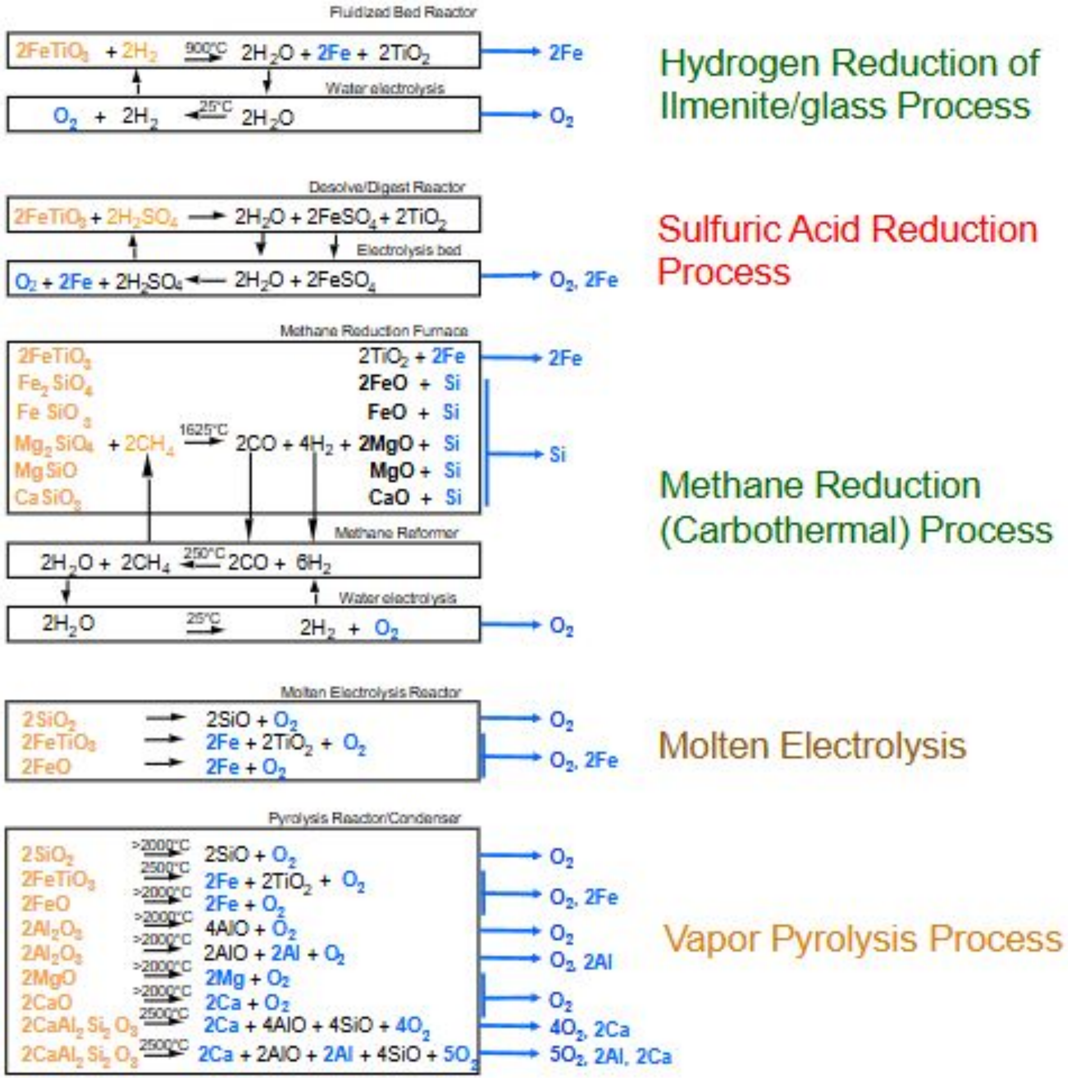
Figure 22: SpaceX Starship Human Landing System

15.) ISRU - Processes

- Molten Electrolysis Reactor: located in the O_2 extraction module, this reactor will perform the $2FeTiO_3 \rightarrow 2Fe+2TiO_2+O_2$ and $2SiO_2 \rightarrow 2SiO+O_2$ reactions (figure 23), extracting oxygen, iron, and titanium dioxide from the Pyroxene, Olivine, and Anorthite minerals abundant in the mare regolith (figure 24).
- Aluminothermic reactor: located within the regolith processing plant, this reactor will perform the $TiO_2 + 4/3Al \rightarrow Ti + 2/3Al_2O_3$ reaction to produce titanium powder for use in 3D printing and $2/3Al_2O_3$.
- Vapor pyrolysis reactor: located within the regolith processing plant, this reactor will perform the $2Al_2O_3 \rightarrow 2AlO+2Al+O_2$ reaction using leftover $2/3Al_2O_3$ from the aluminothermic reactor and pyroxene of the same makeup from the lunar surface to extract aluminum powder for use in 3D printing and oxygen for breathing.

15.) ISRU

Figure 23: Chemical reaction "menu" for lunar mineral ISRU



15.) ISRU

Figure 24: Chemical makeup of various mare regolith minerals

MARE REGOLITH

Ilmenite - 15%

$FeO \cdot TiO_2$	98.5%

Pyroxene - 50%

$CaO \cdot SiO_2$	36.7%
$MgO \cdot SiO_2$	29.2%
$FeO \cdot SiO_2$	17.6%
$Al_2O_3 \cdot SiO_2$	9.6%
$TiO_2 \cdot SiO_2$	6.9%

Olivine - 15%

$2MgO \cdot SiO_2$	56.6%
$2FeO \cdot SiO_2$	42.7%

Anorthite - 20%

$CaO \cdot Al_2O_3 \cdot SiO_2$	97.7%

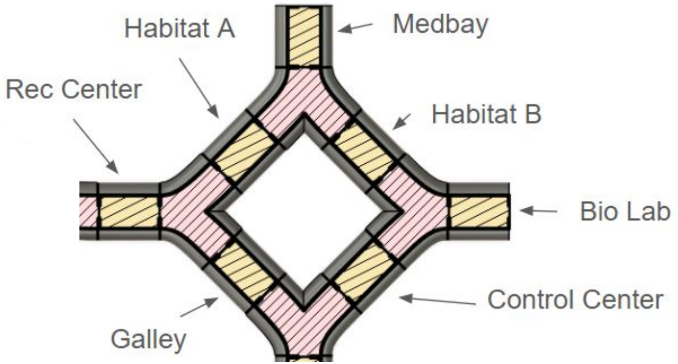
15.) ISRU - Output expectations

- 10000 kg of O_2 would be a realistic production goal of a 2 kW molten electrolysis reactor operating continuously for 6 months, providing some oxidizer to the Starship HLS, and covering all astronaut needs.
- Building supplies (Iron, Titanium, Aluminum) would be produced at a level necessary to construct basic architecture concepts and 3D print spare parts. These materials will help to prove the feasibility of using ISRU for cheap lunar construction material.

16.) Moonbase AndrewMills-1 Final Layout - North Wing

The north wing will be the clean work/living area for the astronauts (figure 25). This will be a shirtsleeve environment, and to pass into the south wing there will be a short cleanroom procedure.

Figure 25: North wing layout



16.) Moonbase AndrewMills-1 Final Layout - West Wing

The west wing will be a clean work area for astronauts, covering the EVA and high water-use modules (figure 26). This will be a shirtsleeve environment, but the water recycler modules would be a high-noise environment, so hearing protection PPE should be worn in and near these areas.

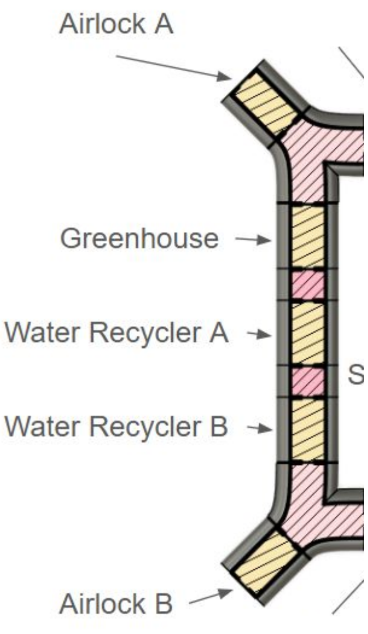
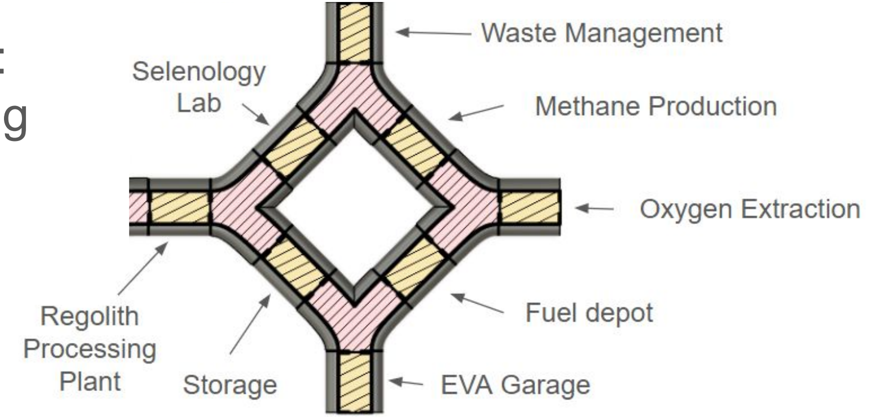


Figure 26: West wing layout

16.) Moonbase AndrewMills-1 Final Layout - South Wing

The south wing will be the dirty work area for the astronauts, covering the entire processing flow of lunar regolith from import to product use (figure 27). Astronauts will have to wear dust mitigation and hearing PPE in the south wing, and go through short cleanroom procedures to enter the north or west wings.

Figure 27: South wing layout

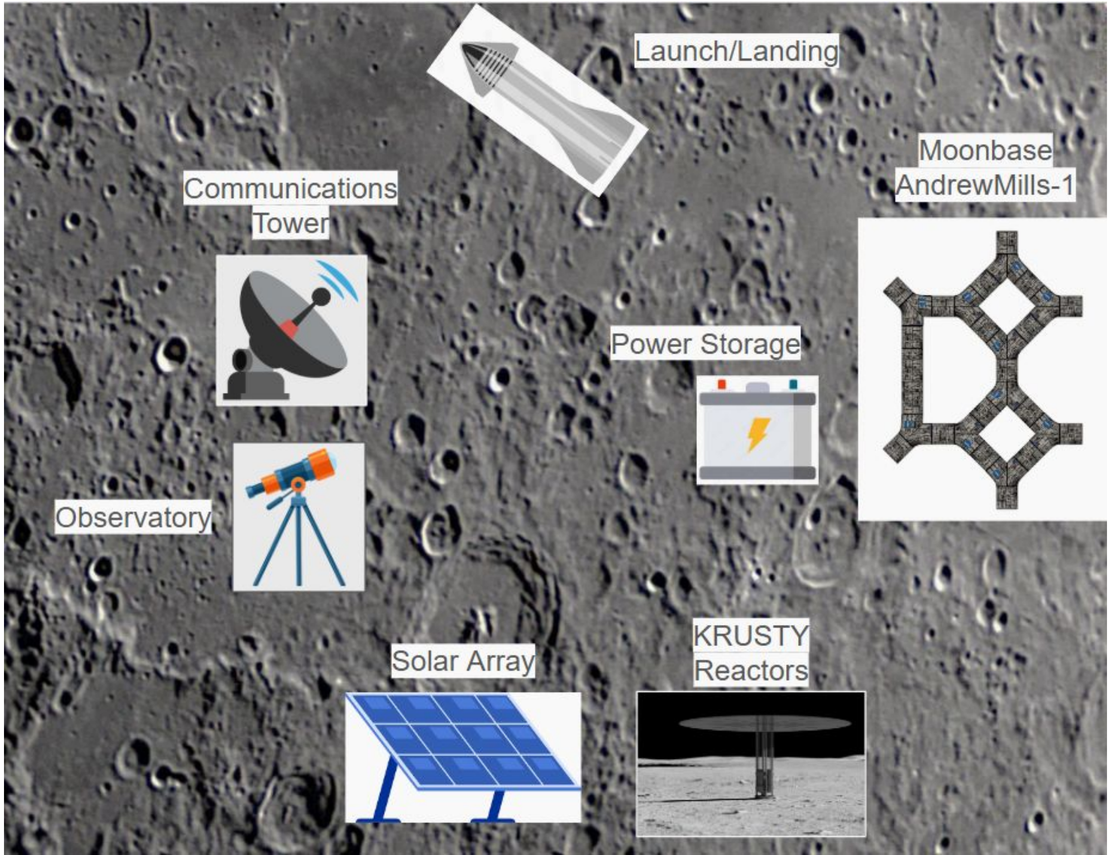


16.) Moonbase AndrewMills-1 Final Layout - Nearby Modules

Some of the modules of the base will be located outside of the pressurized, radiation shielded section, and will therefore be inaccessible without EVA. The layout of these modules is shown in figure 28.

16.) Moonbase AndrewMills-1 Final Layout - Nearby Modules

Figure 28: Layout of the modules nearby to the base



17.) Moonbase AndrewMills-1 Summary and Conclusion

Important things to note about Moonbase AndrewMills-1:

- Located near the equator along the eastern limb of the Moon in Mare Smythii
- Close to large sections of highlands, maria, and young craters
- Always within LOS of Earth
- Regolith shield mitigates temperature, radiation, and meteorite hazards
- Science modules study selenology, agriculture, biology, and astronomy
- ISRU modules will allow astronauts to live off the land in some ways
- A solar array and KRUSTY reactors provide power to the base
- 3 LTVs enable surface mobility across the immediate lunar surface
- SpaceX’s HLS ferries cargo and astronauts to and from the base
- No resupply missions would be required

17.) Moonbase AndrewMills-1 Summary and Conclusion

Moonbase AndrewMills-1 is within the grasp of the technology expected of the next 10 years (2025-2035). With a national effort, this base could be a giant leap in the exploration of our solar system, promising rich scientific research and a vantage point among the stars.

18.) Appendices

- Moon astronomy: Take pictures of the moon as if you are on the moon taking pictures of Earth. Identify moon geographic features.
- Greenhouse simulation: Grow “crops” in a controlled environment, and observe their life cycle.
- Footprint simulation: Choose a lunar regolith simulant and try to recreate the famous Apollo footprints while observing characteristics of the simulant.
- Cratering simulation: Toss garden rocks of various sizes at a tub of lunar regolith simulant and study the ejecta patterns.
- Landing dust plume simulation: Release pressurized air over lunar regolith simulant to test how it would react to a thrust plume. Also tested different strategies for mitigation of dust agitation, such as plastic landing pads or compacted regolith.

18.) Appendices (continued)

- EVA simulation: Don ECLSS gear (heavy clothes, helmet, gloves, backpack) and perform various EVA tasks, such as picking up rocks with tools or stepping through tight doorways.
- Lunar rover activity: Drive an RC car over a bed of regolith simulant, seeing where challenges arise while driving over craters and difficult terrain.
- Regolith deposition activity: Bury an aluminum can in a regolith simulant using different proposed deposition methods to see which methods are easiest.
- Lunar landing and launch piloting: Use a moon landing simulator and choose between telemetry sources you would want in front of you during that landing.